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Correlation Analysis-1

Correlation is a statistical technique that is used to measure and describe the relationship b/w two variables.

- A Correlation requires two scores for each individual.
- These scores normally are identified as x or y . The Pairs of Scores can be listed in a table, or they can be presented graphically in a Scatter Plot.
- The value of a Scatter Plot is that it allows you to see any patterns or trends that exist in the data.
- Application of Correlation:-
 - Correlation is used for Prediction
 - It is used to demonstrate the validity
 - It is used to determine reliability

⇒ Characteristics of relationship:-

1- The direction of relation ship (poistive or negative).

⇒ poistive Correlation, the two variables tends to change in the same direction.

⇒ Negative Correlation: the two variables tends to go in opposite.

2. Form of relationship.

3- The Strength or Consistency of the relationship.

* A Correlation of 1.00 indicates a perfectly consistent relationship whether it is poistive (+1.00) negatively (-1.00).

⇒ Correlation and Caution:-

To establish a Cause and effect relationship, it is necessary to conduct a true experiment in which one variable is manipulated by a researcher and other variables are rigorously controlled.

Although a "strong" correlation exists b/w number of

Churches and Crimes the real
Cause of the relationship is the
size of the Population.

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Correlation Analysis II

⇒ The Pearson Correlation:- The most Common Correlation is the Pearson Correlation (or the Pearson Product-moment Correlation) which measures the degree of straight-line relationship. The Pearson Correlation measures the degree and direction of the linear relationship b/w two variables.

⇒ It is identified by letter r .

⇒ It consist of a ratio comparing

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Correlation Analysis III

⇒ Hypothesis testing with the Pearson Correlation: It is generally

Computed for sample data. A Sample Correlation is used to answer questions about the corresponding Population Correlation.

⇒ The null hypothesis:- is "No" There is no Correlation in the population; or the Population Correlation is zero.

⇒ The alternative hypothesis:- "yes" there is real, non-zero Correlation in the Population.

⇒ The population Correlation is represented by ρ (the Greek letter rho).

Hypothesis would be stated in symbols as.

$H_0: \rho = 0$ (There is no population Correlation)

$H_1: \rho \neq 0$ (There is real Correlation)

⇒ Degrees of Freedom:- The hypothesis test for the Pearson Correlation has degree of freedom defined by $df = n - 2$.

⇒ Correlation in SPSS:-

over a period of
or months; in order to answer
the research question.

→ Longitudinal - Studies:-

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It is useful to organize
the results of the analysis in
one table called an ANOVA
Summary table.

② Tukey's HSD test is the most
commonly used Post hoc test in
Psychological research.

③ One-way ANOVA to understand
whether exam performance
differed based on test anxiety
levels among students into
three independent groups (eg low
medium and high - stressed
students).

④ One-way ANOVA is an

Omnibust test Statistic. eg
understand the entering data
into SPSS through keyboard.

Creating a group variable
called keyboard type (ie) the
independent variable.

ANOVA has the same general
assumptions that apply to all
of the Parametric tests with
an additional assumption of homo-
geneity of variances.

Researchers systematically changes
two or more variables and
then observe how the change
influence another (dependent) variable.
When a research study involves
more than one factor it
is called a factorial design.

ANOVA means groups are defined
by two independent variables.

This technique allows us to
look at the individual and joint
effect of two independent

variables on one dependent variable. The dependent variable for this experiment was the number of errors on the Concept formation tasks.

② one factor is manipulated by the researcher, changing from no-audience to audience, and the second factor is self esteem which varies from high to low.

③ The two factors are used to create a matrix with the different levels of self-esteem defining the rows and the different audience conditions defining the columns.

④ The resulting two-by-two shows for different combinations of the variables, producing four different conditions.

⑤ The dependent variable for the study is the number of errors on the Concept formation task for people observed in four conditions.

(16) In ANOVA, The variable (independent or quasi-independent) that design the groups being compared is called a factor.

(17) Individual Conditions or values that make up a factor are called levels of the factor.

(18) Two-way ANOVA, we calculated three kind of degree of freedom.

(19) The independent-measures ANOVA requires the same three assumptions.

(20) Two Categorical independent variables (e.g. Sex: males/females; age group: young, middle, old)

(21) Two Factor Study examining the depression scores across gender and two treatment conditions. The output gives a plot of the depression scores for males and females across the treatment.

(22) Gender is non-significant and the plot indicates that the mean score of depression for both

- males and females is same.
- 23) The main effect $\bar{e}$ for treatment is based on the tendency for the depression scores to increase from treatment I to treatment II as shown in graph.
 - 24) Treatment I is likely to produce a decrease in depression scores for females than males.
 - 25) Females are more likely to have increased in depression scores after treatment II.
 - 26) ANOVA is Robust test.
 - 27) Each time we do a hypothesis test.
 - 28) we select an alpha level that determines the risk of a type I error.
 - 29) Single experiment requires several hypothesis tests to evaluate all the mean differences.
 - 30) Test wise alpha level simply the alpha level that is selected for each individual hypothesis test.

- 31) Advantage of ANOVA is that it performs all three comparisons simultaneously in one hypothesis test.
- 32) Correlation is a statistical technique that is used to measure and describes the relationship between two variables.

Two variables are simply observed as they exist naturally in the environment - there is no attempt to control or manipulate the variables.

33) Squared Correlation:- measures the proportion of variability in the data that is explained by the relationship b/w x and y . It is sometimes called the Coefficient of determination.

- 34) Correlation is used to demonstrate the validity of the test.
- 35) Validity refers to whether or not the test measures what it claims to measure.

(36) One or two extreme data points often called Outliers.

(37) Dramatic effect on the value of Correlation.

(38) Correlation is a numerical value that describes and measures three characteristics of the relationship b/w x and y .

(39) If variables increase from one individual to another, the y variable also tends to increase, when the x variables decrease, the y variable also decrease.

40) Two variables tends to change in the same direction.

⇒ Notice a relationship b/w the temperature at game time and the bergeres you sell. (Positive)

⇒ On Cold days you sell a lot of Coffee, but Coffee Sales go down as the temperature gose up. (Negative)

(41) x increased by one point, the value of y also changes by a

Consistent and predictable amount
Consistently.

- 92) The Consistency of the relationship is measured by the numerical value of the Correlation.
- 93) Intermediate values b/w 0 and 1 indicate the degree of Consistency.
- 94) Calculating ANOVA, the letter k is used to identify the number of treatment conditions that is the number of levels of the factor.
- 95) One variable is manipulated by a researcher and other variables are rigorously controlled.
- 96) Study shows a positive correlation b/w number of churches and numbers of crimes in towns and cities. It is responsible that small towns would have less crimes and fewer churches and that large cities would have large values for both variables.
- 97) Strong correlation exist b/w no of

Churches and Crime, the real cause of the relationship is the size of the population.

48). Pearson Correlation:- measures the degree and the direction of the linear relationship b/w two variables. There is no Covariability and the resulting correlation is zero.

49) The variability of x is measured by computing SS for the x scores and the variability of y measured by the SS for y scores. (a)

50) The Pearson Correlation is generally computed for sample data.

51) A Sample Correlation is often used to answer questions about the corresponding population correlation.

52) The Pearson Correlation (r) is designed for interval level

(Continuous) variables. It can also be used if you have one continuous variable (e.g. scores on a measure of self-esteem).

- 53) Scale of measurement for the variables should be interval or ratio.
- 54) Dichotomous independent variable (with only two values e.g. gender) and one continuous dependent variable we can still run correlation.
- 55) The null hypothesis is "no". There is no correlation in the population or "the population correlation is zero".
- 56) Alternative hypothesis is "yes". There is real nonzero correlation in the population.
- 57) The population correlation is represented by ρ (the Greek letter rho).
- 58) The size of the value of the correlation coefficient ranges from -1 to 1.
- 59) Analysis were performed to ensure no violation of the assumptions.
- 60) A statistical technique; known as partial correlation. It allows a researcher to measure the relationship

blw two variables while eliminating the influence of a third value.

Pearson Correlation Formula is used with data to form an ordinal scale (ranks) the result is called the Spearman Correlation.

Spearman Correlation. is used to measure the relationship blw x & y when both variables are measured on ordinal scales.

Pearson Correlation measure the degree of linear relationship blw two variables - that is, how well the data points fit on a straight line.

The Spearman (Cor) Correlation measures Consistency, rather than form.

one-directional relationship blw two variables, the relationship is said to be monotonic.

Two values is called a dichotomous variable or a binominal variable. eg
Males versus female, College graduate versus not a college graduate

older than 30 years versus younger than 30 years dicotomous, binomial

67) Correlation b/w the two variables is called the Phi-Coefficient.

68) Phi-Coefficient can be used to assess the relationship b/w two dichotomous variables.

69) Common Statistical Procedure is a Chi-square Statistics.

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One-way ANOVA:- In SPSS The one-way analysis of variance (ANOVA) is used to determine whether there are (many) statistically significant differences b/w the means of two or more independent (unrelated) groups

⇒ Entering Data in SPSS:-

* We separate the groups for analysis by creating a grouping variable is called (keyboard type i.e independent variable) * Number of errors committed is entered under the variable is (errors) (Dependent variable)

- By clicking in the box of the "values" field in the row.
- The data is entered in the data view tab of the SPSS editor.
- A code is entered for categories for independent variable (i.e keyboard A)

(2 keyboard B,) 3 = (keyboard C)
⇒ General assumption include;

- 1- Level of measurement
- 2- Random Sampling
- 3- Independence of Observations.
- 4- Normal distribution.

⇒ Homogeneity of Variance:- It's 1st option gives us the Levene's test. For homogeneity of variances, which tests whether the variance in scores is the same for each of three groups.

⇒ Check the significance value (Sig.) for Levene's test. If this number is greater than 0.05, the assumption is not violated.

⇒ The two tests there (Welch and Brown-Forsythe) are preferable when the assumption of the homogeneity of variance is violated.

⇒ The post-hoc tests table tell us exactly where the differences among the group occurs.

- Post-hoc Comparisons using the Tukey HSD test indicated the no of errors were statistically higher after using the keyboard B ($M = 5.00$, $SD = 2.24$, $p = .018$) and keyboard C ($M = 6.00$, $SD = 1.87$, $p = .004$) as keyboard A ($M = 1.00$, $SD = 1.75$).

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Two-Way ANOVA

- It involving two or more independent variables
- ⇒ when a research study involves more than one factor it is called factorial design.
 - ⇒ when a research study examines ANOVA having exactly two factors. It is called Two-way ANOVA.
 - ⇒ Two-way ANOVA means groups are defined by two independent variables.
 - ⇒ These variables are identified as factor "A" or factor "B".
 - ⇒ The goal of the factorial design is to evaluate the mean differences.

that may be produced by either of the factors acting independently or by the two factors working together
⇒ Main effects and Interactions:-

1. If two variables are moving in the same direction it is called as positive.
2. In ANOVA, the homogeneity of variance assumption will be tested by Levene's Test.
3. In ANOVA, difference exists b/w the treatment condition is termed as B/w-treatments Variance.
4. Correlation is a statistical technique that is used to measure and describe the relationship b/w two variables.
5. Between or within groups are two types of variance which can occur in your data.
6. In ANOVA Welch and Brown-Forsythe test is preferably used when the assumption of the homogeneity of variance is violated.
7. 3 you must have for an ANOVA

to be indicated.

8- The Population Correlation is traditionally represented by the Greek letter ρ .

9- The one-way analysis of variance (ANOVA) is used to determine where there are any statistically significant differences mean of two or more independent groups.

10- For ANOVA, Independent Variable is always Categorical in nature.

11- Variation is not application of Correlation Coefficient.

12- one way ANOVA

2) Assumption of one way ANOVA

2) 2-way ANOVA

2) Assuming 2 way ANOVA

2) ANOVA is Robust test